

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I Priyanga.B(192511102)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Rubrics for Evaluation

<i>Criteria</i>	<i>Weightage</i>	<i>Level 4 – Exemplary</i>	<i>Level 3 – Proficient</i>	<i>Level 2 – Developing</i>	<i>Level 1 – Beginning</i>	<i>Marks(100)</i>
<i>Understanding of Concepts</i>	<i>25%</i>	<i>Demonstrates thorough, accurate understanding of concepts</i>	<i>Good understanding with minor gaps</i>	<i>Basic understanding with some misconceptions</i>	<i>Poor understanding; major misconceptions</i>	
<i>Explanation</i>	<i>25%</i>	<i>Detailed, well-explained answers with relevant examples</i>	<i>Clear explanation with adequate details</i>	<i>Limited explanation; lacks depth</i>	<i>Poor or unclear explanation</i>	
<i>Application</i>	<i>20%</i>	<i>Effectively applies concepts with strong analytical ability</i>	<i>Applies concepts with minor errors</i>	<i>Limited application with weak analysis</i>	<i>No application or incorrect analysis</i>	
<i>Organization</i>	<i>15%</i>	<i>Well-structured answers with logical flow and coherence</i>	<i>Organized with minor issues in flow</i>	<i>Basic structure, lacks clarity</i>	<i>Poor organization, incoherent answers</i>	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image appears too bright.

Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

Technique Applied: Power-Law (Gamma) Transformation with $\gamma > 1$

Concept

A grey level transformation maps every input intensity r of an image to an output intensity s using a function $s = T(r)$, applied independently to each pixel. When an image is too bright, most pixel values are concentrated in the higher end of the intensity range (close to 255), which compresses visual detail in bright regions and reduces contrast. The Power-Law (Gamma) transformation, given by $s = c \cdot r^\gamma$, is the most suitable choice here. Choosing $\gamma > 1$ compresses the higher intensity values and expands the darker range, effectively darkening the overall image while pulling back detail that was washed out by excessive brightness.

Application / Working Process

The transformation is applied by first normalizing pixel intensities to the range $[0,1]$, raising each value to the power γ (a value greater than 1, e.g., 1.5–2.5, chosen experimentally or based on the image histogram), and multiplying by a scaling constant c (usually 1) before mapping back to $[0,255]$. Since the function is non-linear, brighter pixels are pushed down more strongly than mid-tone pixels, which preserves the relative structure of the image while correcting the overexposure.

Expected Outcome

The processed image shows improved visibility of details that were earlier lost in the bright regions (e.g., highlights, sky, or over-illuminated areas). The overall dynamic range becomes better balanced, contrast is enhanced, and the image looks visually corrected rather than washed out, without introducing noise or artifacts.

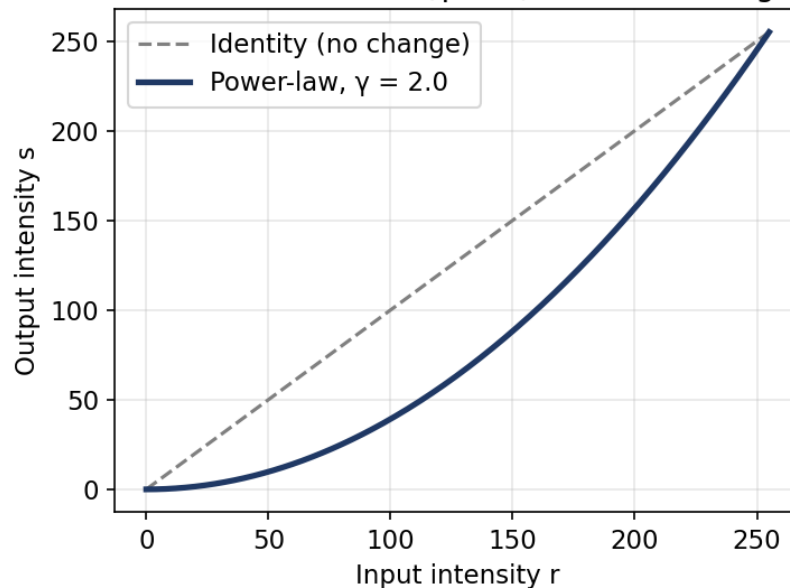
Worked Example / Diagram

Take a pixel with input intensity $r = 200$ (bright region), scaled to $[0,1]$ as 0.784, with $\gamma = 2$:

$$s = 255 \times (200/255)^2 = 255 \times 0.615 = 157$$

The bright pixel (200) is pulled down to 157, compressing the bright range and revealing detail that was clipped near white.

Power-Law Transformation ($\gamma > 1$) — darkens bright pixels



2.A dark image lacks detail.

Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

Technique Applied: Log Transformation (alternatively, Power-Law with $\gamma < 1$)

Concept

The Log transformation is defined as $s = c \cdot \log(1 + r)$, where r is the input intensity, s is the output intensity, and c is a scaling constant. This function has a characteristic shape that expands the range of low (dark) intensity values while compressing the range of high (bright) intensity values. For a dark image, most pixel values are clustered near 0, and the human eye finds it difficult to distinguish subtle differences within that narrow dark band. The logarithmic curve stretches this narrow dark band into a much wider range of output values, making previously indistinguishable dark details visible.

Application / Working Process

Each pixel intensity is transformed using $s = c \cdot \log(1 + r)$, with c typically chosen as $c = 255 / \log(1 + 255)$ so that the output remains within the valid $[0, 255]$ range. Because the log function grows quickly for small r and slowly for large r , dark pixels receive a proportionally larger boost in intensity compared to already-bright pixels, which stay relatively unchanged.

Expected Outcome

Hidden structures and textures in the shadow regions of the image become clearly visible. The overall brightness of the dark image increases in a non-linear, perceptually meaningful way, and

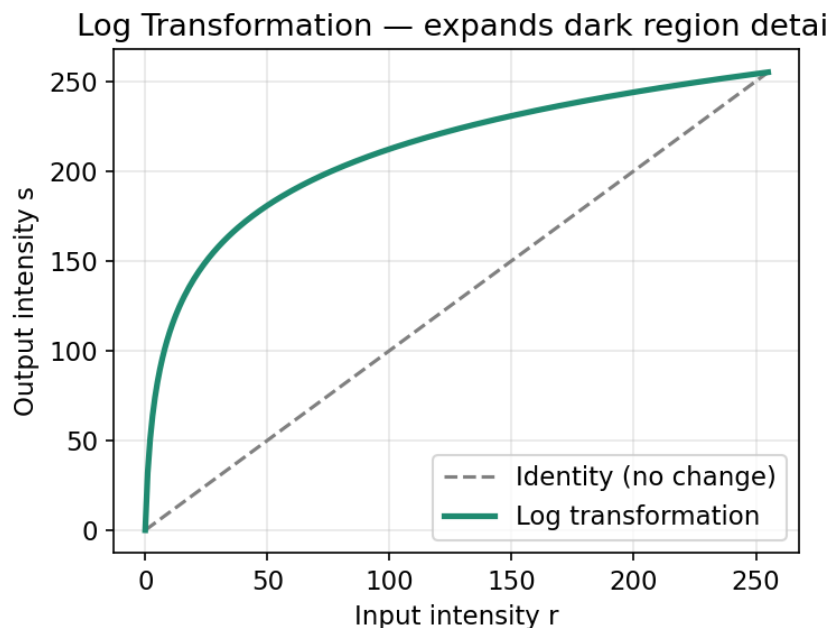
the enhanced image reveals detail without over-brightening the parts of the image that were already reasonably visible.

Worked Example / Diagram

Take a dark pixel with input intensity $r = 10$, using $c = 255/\log(256) \approx 46$:

$$s = 46 \times \log(1 + 10) = 46 \times 2.398 = 110$$

The dark pixel (10) is boosted to 110 — a large jump — while a bright pixel like $r = 250$ would only move to ≈ 253 , confirming the curve mainly stretches dark tones.



3. An image has poor contrast.

Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

Technique Applied: Histogram Equalization

Concept

Histogram Equalization (HE) is a histogram-based contrast enhancement technique that redistributes the intensity values of an image so that the output histogram is as close to a uniform (flat) distribution as possible. A poor-contrast image typically has a histogram where pixel intensities are clustered within a narrow range, causing the image to appear flat, dull, or hazy. HE works on the reasoning that spreading out the most frequent intensity values over the entire available range increases the overall contrast of the image.

Application / Working Process

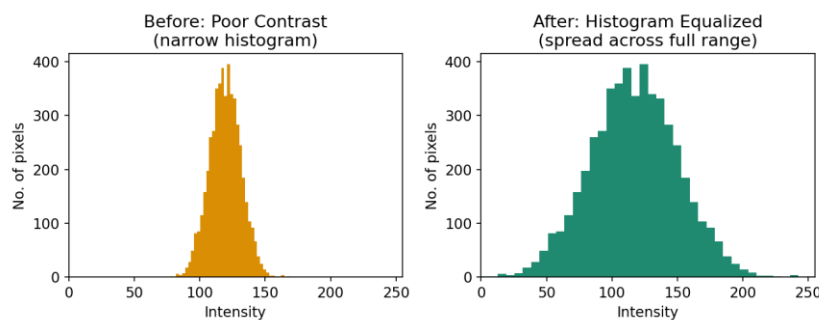
The technique computes the histogram of the image, then derives the Cumulative Distribution Function (CDF) of the intensity levels. Each pixel's original intensity is then mapped to a new intensity using the transformation $s = (L-1) \cdot \text{CDF}(r)$, where L is the number of grey levels (256 for 8-bit images). Intensity values that occur frequently are spread further apart in the output, effectively stretching the histogram across the full dynamic range.

Expected Outcome

The resulting image shows a broader spread of intensity values, revealing details and structures that were previously indistinguishable due to poor contrast. The output histogram becomes flatter and closer to uniform, and the visual sharpness and clarity of the image improve significantly, especially in regions that were originally washed out or muddy.

Worked Example / Diagram

Before equalization the histogram is narrow (pixels bunched around mid-grey $\Rightarrow$ low contrast). After equalization the same pixels are remapped using the CDF and spread across the full 0–255 range:



4.A low-contrast image requires uniform intensity distribution.

Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

Technique Applied: Histogram Equalization

Concept

Since the requirement is specifically a uniform intensity distribution, Histogram Equalization is the exact and most justified method, because its underlying goal is precisely to transform the intensity distribution of an image so that it approximates a uniform histogram. The method treats each grey level's cumulative probability as an indicator of how much 'space' it should occupy in the new, stretched intensity scale.

Application / Working Process

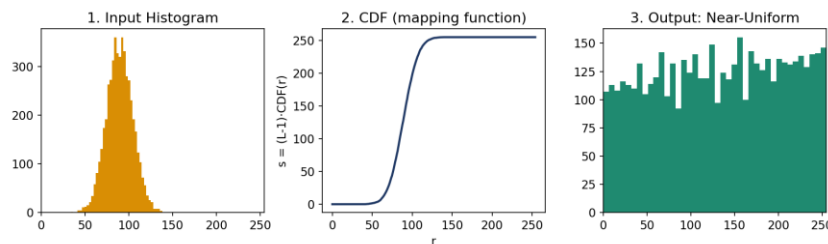
The method operates in three stages: (1) compute the normalized histogram (probability of occurrence) of each intensity level in the image; (2) calculate the cumulative distribution function (CDF) from this histogram; (3) use the CDF, scaled by $(L-1)$, as the transformation function that maps every old intensity level r to a new intensity level s . Because the CDF is a monotonically non-decreasing function, this mapping preserves the relative order of intensities while equalizing their spread.

Expected Outcome

After equalization, intensity levels that were previously bunched together are spread across a wider range, and levels with high frequency get proportionally more of the output range assigned to them. This produces an image with enhanced global contrast and an intensity histogram that is much closer to uniform than the original, satisfying the stated requirement directly.

Worked Example / Diagram

The three-stage pipeline — input histogram → cumulative distribution function (CDF) used as the mapping → resulting near-uniform output histogram:



5. An image shows uneven brightness across regions.

Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

Technique Applied: Adaptive Histogram Equalization (AHE / CLAHE)

Concept

Uneven brightness across an image, also known as non-uniform illumination, cannot be corrected effectively by a single global transformation (like standard Histogram Equalization) because a global mapping applies the same correction everywhere, regardless of local lighting conditions. Adaptive Histogram Equalization (AHE), and its improved variant Contrast Limited Adaptive Histogram Equalization (CLAHE), address this by computing and applying histogram equalization locally, within small regions (tiles) of the image, rather than globally.

Application / Working Process

The image is divided into small, non-overlapping tiles (e.g., 8×8 blocks). Histogram equalization

is performed independently within each tile, so each local region gets its own contrast correction based on its own intensity distribution. To avoid over-amplifying noise in nearly uniform regions, CLAHE clips the histogram at a predefined limit before computing the CDF. Finally, bilinear interpolation is applied at tile boundaries to eliminate visible seams between adjacent tiles.

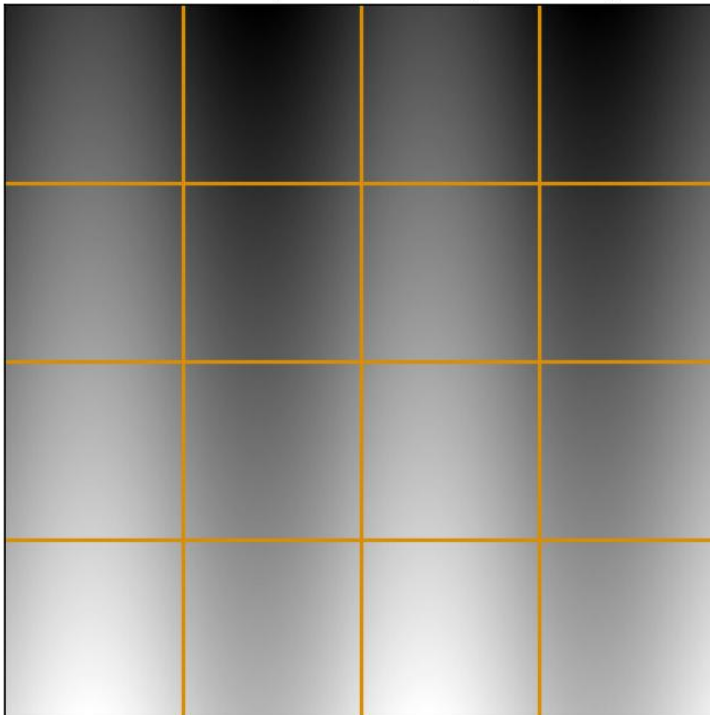
Expected Outcome

Regions that were too dark or too bright due to uneven illumination are individually corrected, resulting in a visually balanced image where local details are enhanced uniformly across all regions. Unlike global histogram equalization, this method avoids over-enhancing already well-lit areas while still correcting poorly-lit ones, producing a more natural-looking, evenly illuminated output.

Worked Example / Diagram

The image is split into tiles (e.g., 4×4 blocks shown below); each tile is histogram-equalized using only its own local pixel values, so a dark tile and a bright tile each get corrected independently:

Image Divided into Local Tiles (CLAHE)
Each tile equalized independently



6. A noisy image contains random intensity variations.

Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

Technique Applied: Mean (Averaging) Filter

Concept

Random intensity variations across an image (typically resembling Gaussian noise) can be reduced using spatial smoothing filters, which work on the principle that noise is a high-frequency, random component that differs from pixel to pixel, whereas real image content changes more gradually. The Mean (Averaging) filter is a linear spatial filter that replaces each pixel's value with the average intensity of its neighborhood, thereby smoothing out sudden random fluctuations.

Application / Working Process

A mask (commonly 3×3 or 5×5) of equal weights (e.g., all $1/9$ for a 3×3 mask) is convolved with the image. For every pixel, the filter computes the arithmetic mean of the intensities within the neighborhood defined by the mask and assigns this average as the new value of the center pixel. This process is repeated across the entire image.

Expected Outcome

Random noise fluctuations, which tend to average out toward the local mean, are significantly reduced, producing a smoother image. However, because averaging is applied uniformly, fine details and edges are also blurred to some extent — a trade-off that is acceptable when the primary goal is noise suppression rather than detail preservation.

Worked Example / Diagram

3×3 neighborhood of a noisy pixel (center = 150, a random spike) and the Mean filter mask:

52	55	50
54	150	53
51	56	55

$$\text{New center value} = (52+55+50+54+150+53+51+56+55) / 9 = 576/9 \approx 64$$

The noisy spike (150) is diluted into the average, reducing its impact but slightly softening the surrounding detail.

7. An image requires noise reduction without significant loss of details.

Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

Technique Applied: Median Filter

Concept

When detail preservation is important alongside noise reduction, the Median Filter is the most appropriate choice. It is a non-linear spatial filter that replaces each pixel's value with the median of the intensity values in its neighborhood, rather than the mean. Because the median is robust to outliers (extreme values such as salt-and-pepper noise), it removes noisy pixels without significantly blurring edges and fine structures, unlike the mean filter.

Application / Working Process

A neighborhood window (commonly 3×3) is placed over each pixel. All the pixel intensities within this window are sorted in ascending order, and the middle (median) value is selected as the new intensity for the center pixel. This process is repeated for every pixel in the image, sliding the window across the entire image.

Expected Outcome

Impulse-type noise (isolated bright or dark pixels) is effectively eliminated because such extreme values are pushed to the ends of the sorted list and are unlikely to become the median. At the same time, edges and structural details are largely preserved because the median filter does not blend pixel values the way averaging does, resulting in a cleaner image with minimal loss of sharpness.

Worked Example / Diagram

Same noisy 3×3 neighborhood, now processed with the Median filter:

52	55	50
54	150	53
51	56	55

Sorted values: 50, 51, 52, 53, 54, 55, 55, 56, 150 → Median (5th value) = 54

The outlier 150 is completely discarded (it is only used to determine ranking), so the noise is removed while the true local intensity (~ 54) is preserved exactly — unlike the blurred result from the mean filter.

8. An image appears blurred after smoothing.

Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

Technique Applied: Laplacian Sharpening (Second-Order Derivative Filter)

Concept

Smoothing operations reduce high-frequency content in an image, which includes both noise and important edge/detail information, resulting in blur. To restore lost details, a sharpening filter based on the Laplacian operator is used, since it is a second-order derivative filter that highlights regions of rapid intensity change (edges, fine details), which are exactly the components that smoothing suppresses.

Application / Working Process

The Laplacian operator is computed using a mask such as $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ (or its 8-neighbor variant), which is convolved with the image to compute the second derivative at every pixel. This Laplacian image is then subtracted from (or added to, depending on the mask's center sign convention) the original blurred image using the formula $g(x,y) = f(x,y) - \nabla^2 f(x,y)$, which reintroduces the high-frequency edge information into the image.

Expected Outcome

The resulting sharpened image shows enhanced edges and restored fine detail, with a visibly crisper appearance compared to the blurred input. Since the Laplacian is sensitive to noise, care must be taken (e.g., mild pre-smoothing) if the original blur was caused by noise-suppression filtering, to avoid amplifying residual noise along with the edges.

Worked Example / Diagram

Laplacian kernel (4-neighbor form) used to compute the second derivative:

0	1	0
1	-4	1
0	1	0

Take an edge pixel: center intensity = 100, with the four neighbors (up, down, left, right) all = 60 (a dark region next to a brighter pixel):

$$\nabla^2 f = (60+60+60+60) - 4(100) = 240 - 400 = -160$$

Sharpened output: $g = f - \nabla^2 f = 100 - (-160) = 260 \rightarrow$ clipped to 255

The center pixel, already brighter than its surroundings, is pushed even further apart from its neighbors, which visibly sharpens that edge boundary.

9. Fine details in an image are not clearly visible.

Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

Technique Applied: High-Pass (High-Boost) Filtering

Concept

Fine details in an image correspond to high-frequency components in the spatial/frequency domain, since they involve rapid, small-scale changes in intensity. A High-Pass filter is designed specifically to allow these high-frequency components to pass through while attenuating low-frequency components (smooth, slowly-varying regions), thereby emphasizing detail. A related and often more practical variant is High-Boost filtering, which amplifies high frequencies while still retaining some of the low-frequency background.

Application / Working Process

High-pass filtering can be implemented in the spatial domain using a mask that emphasizes the center pixel relative to its neighbors (a sharpening-type kernel), or in the frequency domain by multiplying the image's Fourier transform with a high-pass transfer function (e.g., an Ideal, Butterworth, or Gaussian high-pass filter) and taking the inverse transform. High-boost filtering is computed as: $\text{High-boost} = A \cdot f(x,y) - \text{Low-pass}(f(x,y))$, where $A \geq 1$ controls how much of the original image is retained alongside the enhanced high frequencies.

Expected Outcome

Edges, textures, and fine structural details that were previously subdued become significantly more visible and well-defined. Because high-frequency components are selectively boosted rather than uniformly amplified, the technique enhances detail without introducing the same level of noise amplification as naive sharpening.

Worked Example / Diagram

High-boost kernel ($A = 1.5$, sharpens while retaining background):

0	-1	0
-1	6.5	-1
0	-1	0

This kernel is built from $\text{High-boost} = A \cdot f - \text{Low-pass}(f)$: the center weight rises above the pure high-pass value (5) to $4A+1 = 4(1.5)+1 \approx 6.5-7$ as A increases, while the four neighbor weights stay at -1 .

A higher A keeps more of the original image; A close to 1 gives pure high-pass output (edges only, background suppressed), while a larger A retains more of the background brightness alongside the sharpened fine detail.

10.A noisy image must be preprocessed before further analysis.

Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

Technique Applied: Gaussian Smoothing Filter (spatial domain preprocessing)

Concept

Before further analysis (such as segmentation, edge detection, or feature extraction), it is standard practice to preprocess a noisy image using a spatial smoothing filter that reduces noise while preserving essential structural information as much as possible. The Gaussian filter is well suited for this because its weights are derived from a Gaussian (bell-shaped) function, giving higher importance to pixels closer to the center of the neighborhood and progressively less weight to pixels farther away, which produces smoother and more natural noise suppression than a simple mean filter.

Application / Working Process

A Gaussian kernel is generated using the formula $G(x,y) = (1/2\pi\sigma^2) \cdot e^{-((x^2+y^2)/2\sigma^2)}$, where σ (standard deviation) controls the degree of smoothing — a larger σ smooths more aggressively. This kernel is convolved with the noisy image, so each output pixel becomes a weighted average of its neighborhood, with weights following the Gaussian distribution.

Expected Outcome

The preprocessed image has significantly reduced random noise while retaining a reasonable degree of edge and structural information (better than a plain mean filter), making it far more suitable as an input for subsequent analysis stages such as segmentation or feature detection, where noise could otherwise cause spurious or unreliable results.

Worked Example / Diagram

5×5-style behavior approximated with a 3×3 Gaussian kernel ($\sigma \approx 1$), normalized by 16:

1	2	1
2	4	2
1	2	1

For neighborhood [[10,12,11],[13,90,14],[9,11,10]] (90 = noise spike):

$$\begin{aligned} \text{New value} &= (1 \cdot 10 + 2 \cdot 12 + 1 \cdot 11 + 2 \cdot 13 + 4 \cdot 90 + 2 \cdot 14 + 1 \cdot 9 + 2 \cdot 11 + 1 \cdot 10) / 16 = \\ &= (10 + 24 + 11 + 26 + 360 + 28 + 9 + 22 + 10) / 16 = 500 / 16 \approx 31 \end{aligned}$$

The center-weighted average softens the noise spike (90→31) more gently than a plain mean filter would, while still respecting nearby structure more than a uniform average.